

Sampling

Signals and Linear Systems

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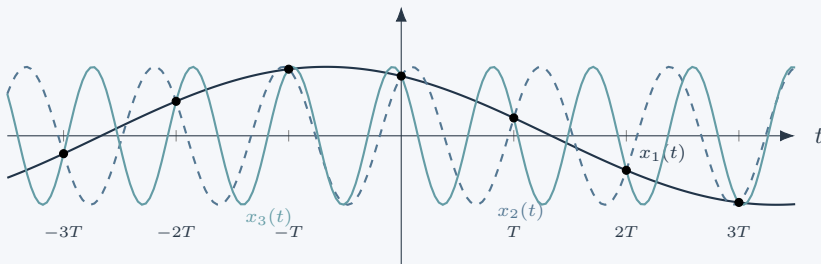
- **Introduction:** Representation of a Continuous-Time Signal by Its Samples
- **Impulse-Train Sampling:** Mathematical Model and Derivation
- **The Sampling Theorem:** Nyquist Rate and Aliasing
- **Signal Reconstruction:**
 - Ideal Low-Pass Filtering
 - Zero-Order Hold (ZOH)
 - Linear Interpolation
- **Revisiting the DFT:** Practical considerations and periodic signals

Introduction and Terminology

- In our review of the DFT, we established that evaluating a continuous-time signal $x(t)$ at uniform intervals T yields a discrete sequence $x[n] = x(nT)$.
- **Question:** Under what conditions can a continuous-time signal be uniquely determined by and exactly recovered from its discrete samples?
- This exact recovery is possible if the signal is **band-limited** (its Fourier Transform is zero outside a finite band of frequencies) and if the samples are taken sufficiently close together.
- This fundamental concept is known as the **Nyquist-Shannon Sampling Theorem**.

Ambiguity in Representation by Samples

- Generally, infinitely many continuous-time signals can yield the exactly same discrete sequence when sampled at interval T .

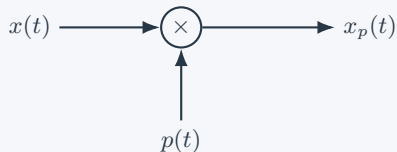


- To analyze sampling mathematically, it is convenient to represent the sampling process as multiplying the continuous signal $x(t)$ by a periodic impulse train $p(t)$.

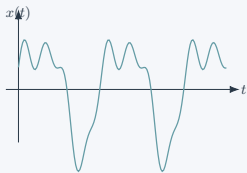
Impulse-Train Modulator

$$p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT)$$

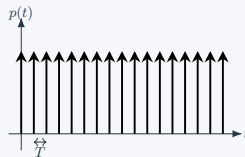
$$x_p(t) = x(t)p(t) = \sum_{n=-\infty}^{+\infty} x(nT)\delta(t - nT)$$



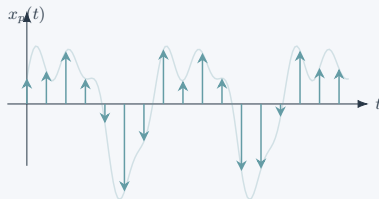
Visualizing Impulse-Train Sampling



Continuous signal $x(t)$.



Impulse train $p(t)$.



Sampled signal $x_p(t)$.

Derivation of $X_p(j\omega)$

- To see the effect of sampling, we need the Fourier Transform of $x_p(t) = x(t)p(t)$.
- Let's briefly derive the **duality principle**. From the inverse Fourier Transform:

$$x(t) = \frac{1}{2\pi} \int X(j\omega) e^{j\omega t} d\omega \implies 2\pi x(-t) = \int X(j\omega) e^{-j\omega t} d\omega$$

- Swapping t and ω :

$$2\pi x(-\omega) = \int X(jt) e^{-j\omega t} dt = \mathcal{F}\{X(jt)\}$$

- Hence, the duality principle is as follows:

$$x(t) \xleftrightarrow{\mathcal{F}} X(j\omega) \implies X(jt) \xleftrightarrow{\mathcal{F}} 2\pi x(-\omega)$$

Multiplication is Convolution in the Frequency Domain

- We know from the Convolution Property that,

$$x_1(t) * x_2(t) \xleftrightarrow{\mathcal{F}} X_1(j\omega)X_2(j\omega)$$

- By duality,

$$X_1(jt) * X_2(jt) \xleftrightarrow{\mathcal{F}} 2\pi x_1(-\omega)x_2(-\omega)$$

$$X_1(jt) \xleftrightarrow{\mathcal{F}} 2\pi x_1(-\omega)$$

$$X_2(jt) \xleftrightarrow{\mathcal{F}} 2\pi x_2(-\omega)$$

- Substituting $X_1(jt)$ with $u(t)$ and $X_2(jt)$ with $v(t)$:

$$u(t) \xleftrightarrow{\mathcal{F}} 2\pi x_1(-\omega) = U(j\omega)$$

$$v(t) \xleftrightarrow{\mathcal{F}} 2\pi x_2(-\omega) = V(j\omega)$$

- Substituting we get:

$$u(t)v(t) \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} U(j\omega) * V(j\omega)$$

- Hence, we can write the Fourier Transform of the sampled signal as:

$$X_p(j\omega) = \frac{1}{2\pi} [X(j\omega) * P(j\omega)]$$

- Directly taking the Fourier Transform results in a summation that is hard to work with. Instead, we can use the Fourier Series.
- $p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT)$ is clearly periodic with period T and thus using Fourier Series:

$$p(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_s t}, \quad \text{where } \omega_s = \frac{2\pi}{T}.$$

- Using the **sifting property** $\int f(t)\delta(t)dt = f(0)$ on period $[-T/2, T/2]$:

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_s t} dt = \frac{1}{T}$$

- Hence, we can write:

$$p(t) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} e^{jk\omega_s t}$$

- We know the FT of an impulse is 1. By duality, the FT of a constant is an impulse.

$$\delta(t) \xleftrightarrow{\mathcal{F}} 1$$
$$1 \xleftrightarrow{\mathcal{F}} 2\pi\delta(\omega)$$

- Using the phase-shifting property:

$$e^{j\omega_0 t} x(t) \xleftrightarrow{\mathcal{F}} \int_{-\infty}^{+\infty} x(t) e^{-j(\omega - \omega_0)t} dt = X(j(\omega - \omega_0))$$

- Thus, we have:

$$e^{jk\omega_s t} \xleftrightarrow{\mathcal{F}} 2\pi\delta(\omega - k\omega_s)$$

- Taking the Continuous-Time Fourier Transform of $p(t)$ term-by-term yields another impulse train:

$$P(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} \mathcal{F}\{e^{jk\omega_s t}\} = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s)$$

Spectrum of the Sampled Signal

- Now we evaluate the convolution $X_p(j\omega) = \frac{1}{2\pi} X(j\omega) * P(j\omega)$:

$$\begin{aligned} X_p(j\omega) &= \frac{1}{2\pi} X(j\omega) * \left[\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s) \right] \\ &= \frac{1}{T} \sum_{k=-\infty}^{+\infty} X(j\omega) * \delta(\omega - k\omega_s) \end{aligned}$$

- For convolution with a shifted impulse, we use the integral definition and sifting property:

$$X(j\omega) * \delta(\omega - \omega_0) = \int_{-\infty}^{+\infty} X(j\theta) \delta(\omega - \theta - \omega_0) d\theta = X(j(\omega - \omega_0))$$

- Thus, we obtain the fundamental equation of sampling:

Spectrum of Sampled Signal

$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X(j(\omega - k\omega_s))$$

- $X_p(j\omega)$ consists of periodically repeated copies of $X(j\omega)$, scaled by $1/T$.

The Sampling Theorem

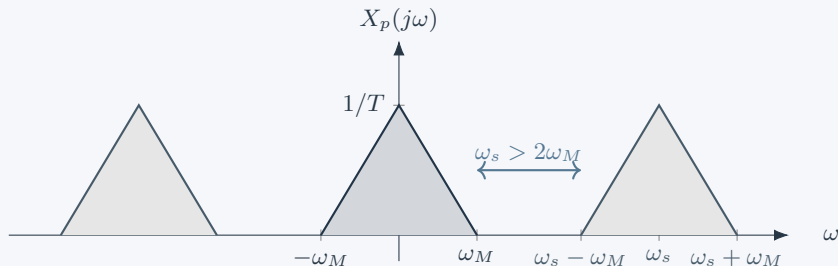
- Suppose $x(t)$ is a **band-limited** signal, meaning $X(j\omega) = 0$ for $|\omega| > \omega_M$.
- The copies of $X(j\omega)$ in $X_p(j\omega)$ are centered at integer multiples of ω_s .
- To avoid overlap between adjacent copies, the separation ω_s must be strictly greater than twice the maximum frequency ω_M .

Nyquist-Shannon Sampling Theorem

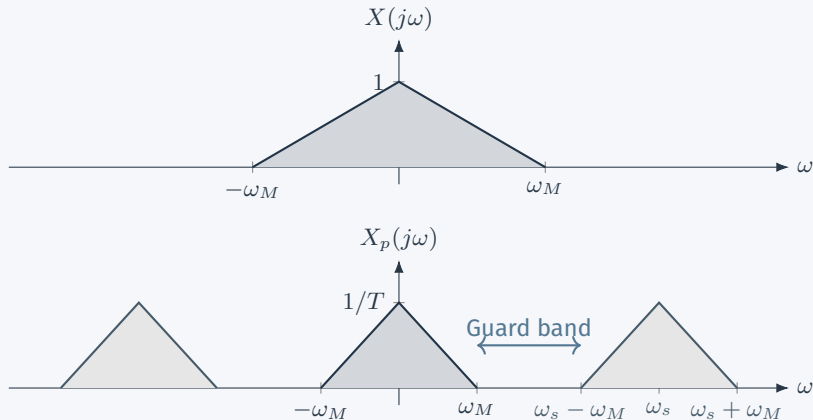
Let $x(t)$ be a band-limited signal with $X(j\omega) = 0$ for $|\omega| > \omega_M$. Then $x(t)$ is uniquely determined by its samples $x(nT)$, $n = 0, \pm 1, \pm 2, \dots$, if

$$\omega_s > 2\omega_M$$

where $\omega_s = 2\pi/T$. The minimum sampling frequency $2\omega_M$ is called the **Nyquist rate**.

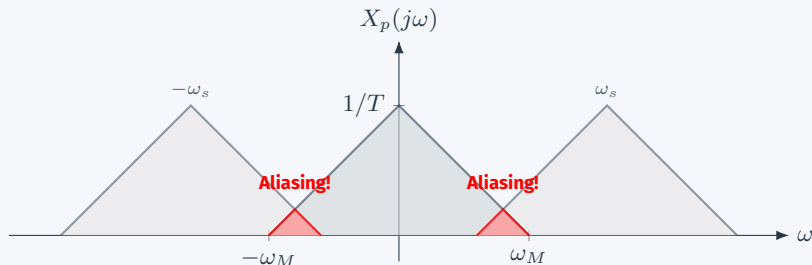


Visualizing the Spectrum: Oversampling



- When $\omega_s > 2\omega_M$, exact copies of the original spectrum are perfectly separated.
- The original signal can be recovered by isolating the baseband copy using a low-pass filter!

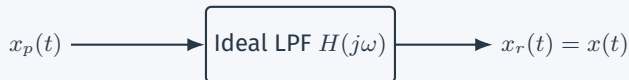
Visualizing the Spectrum: Undersampling (Aliasing)



- When $\omega_s < 2\omega_M$, the shifted copies overlap with the baseband spectrum.
- High frequencies fold over into the low-frequency band, distorting the signal. This phenomenon is called **aliasing**.
- The original signal $x(t)$ cannot be recovered by low-pass filtering.

Signal Reconstruction

- By the sampling theorem, if $\omega_s > 2\omega_M$, the **baseband spectrum** of $x_p(t)$ (the copy of the spectrum $X(j\omega)$ at the origin) is an exact replica of $X(j\omega)$ scaled by $1/T$.
- We can recover $x(t)$ by applying an **ideal low-pass filter** to $x_p(t)$.

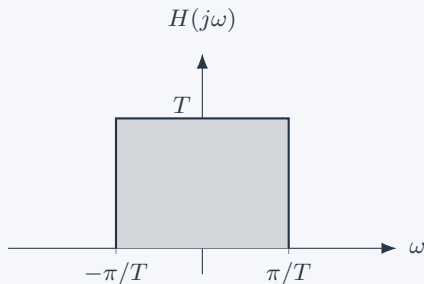


- The frequency response of the ideal LPF must be:

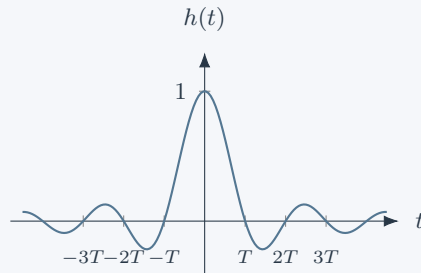
$$H(j\omega) = \begin{cases} T & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases} \quad \text{where} \quad \omega_M < \omega_c < \omega_s - \omega_M$$

- Commonly, we choose the cutoff frequency exactly in the middle: $\omega_c = \omega_s/2$, because it follows the assumption that $\omega_M < \omega_s/2$.

The Ideal Low-Pass Filter



Frequency Response $H(j\omega)$



Impulse Response $h(t)$

- The frequency response $H(j\omega)$ is a rectangular pulse with cutoff $\omega_c = \omega_s/2 = 2\pi f_s/2 = \pi/T$.
- The time-domain impulse response $h(t) = \text{sinc}(t/T)$ extends infinitely in both directions, making the ideal LPF **unrealizable**.

To find the time-domain impulse response $h(t)$ of an ideal low-pass filter with cutoff $\omega_c = \pi/T$ and constant amplitude T , we compute the Inverse Fourier Transform:

$$\begin{aligned} h(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} H(j\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\pi/T}^{\pi/T} T e^{j\omega t} d\omega \\ &= \frac{T}{2\pi} \left[\frac{e^{j\omega t}}{jt} \right]_{-\pi/T}^{\pi/T} = \frac{T}{\pi t} \left(\frac{e^{j(\pi/T)t} - e^{-j(\pi/T)t}}{2j} \right) \\ &= \frac{T}{\pi t} \sin\left(\frac{\pi t}{T}\right) = \text{sinc}\left(\frac{t}{T}\right) \quad \left[\text{Recall } \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x} \right] \end{aligned}$$

Result: The time-domain filter corresponding to a perfect frequency cutoff is exactly the infinitely wide sinc function!

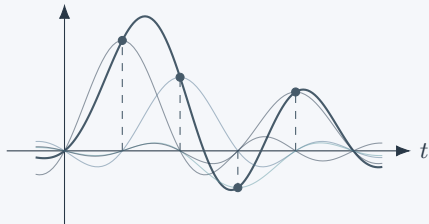
- The impulse response $h(t)$ of the ideal LPF (cutoff $\omega_c = \pi/T$) is:

$$h(t) = \text{sinc}\left(\frac{t}{T}\right) = \frac{\sin(\pi t/T)}{\pi t/T}$$

- The reconstructed signal $x_r(t) = x_p(t) * h(t)$ becomes:

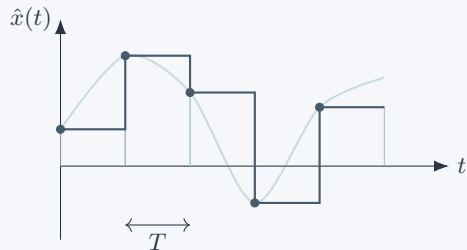
$$\begin{aligned} x_r(t) &= \left(\sum_{n=-\infty}^{+\infty} x(nT) \delta(t - nT) \right) * h(t) \\ &= \sum_{n=-\infty}^{+\infty} x(nT) \text{sinc}\left(\frac{t - nT}{T}\right) \end{aligned}$$

- Superposition of shifted sinc functions.
- However**, $h(t)$ is non-causal and decays very slowly. It is **unrealizable** in practice. We need approximations.



The simplest practical reconstruction method is the **Zero-Order Hold (ZOH)**.

- **Concept:** Simply hold the sample value constant until the next sample arrives.
- This forms a “staircase” approximation $x_0(t)$ of the continuous signal.
- Widely used in Digital-to-Analog Converters (DACs).



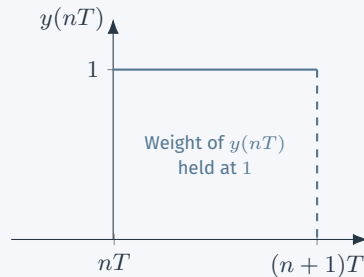
- We can view this mathematically as filtering the impulse-sampled signal $x_p(t)$ with a rectangular pulse $h_0(t)$.

Why is $h_0(t)$ a Rectangle?

- In a reconstructed output signal, an original sample $y(nT)$ contributes strictly via a shifted, scaled version of the base filter: $y(nT) \cdot h(t - nT)$.
- The shape of $h(t)$ dictates the **weighting** of the sample $y(nT)$ over continuous time.

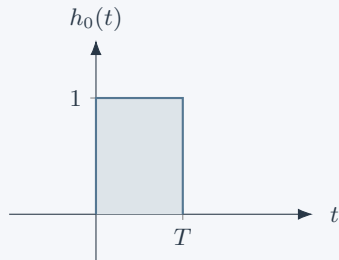
For Zero-Order Hold:

- We desire the sample $y(nT)$ to simply hold its full weight of 1 constant precisely until the next sample at $(n + 1)T$ arrives.
- Thus, the isolated weight of $y(nT)$ must jump to 1 at $t = nT$ and abruptly drop back to 0 at $t = (n + 1)T$.
- Therefore, the underlying impulse response $h_0(t)$ **must** inherently be a rectangular block formally spanning from 0 to T !

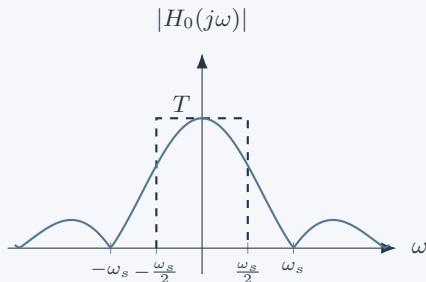


The ZOH impulse response $h_0(t)$ is simply a rectangular pulse equal to 1 over $[0, T]$. We construct $H_0(j\omega)$ strictly via continuous Fourier integration:

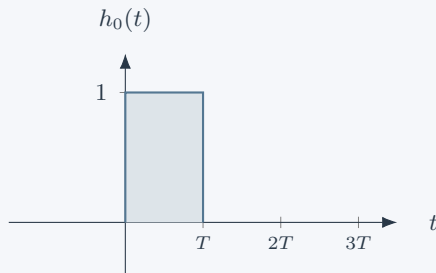
$$\begin{aligned} H_0(j\omega) &= \int_{-\infty}^{\infty} h_0(t) e^{-j\omega t} dt \\ &= \int_0^T (1) e^{-j\omega t} dt = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^T \\ &= \frac{1 - e^{-j\omega T}}{j\omega} = \frac{e^{-j\omega T/2} (e^{j\omega T/2} - e^{-j\omega T/2})}{j\omega} \\ &= e^{-j\omega T/2} \left[\frac{2}{\omega} \sin \left(\frac{\omega T}{2} \right) \right] \\ &= T e^{-j\omega T/2} \operatorname{sinc} \left(\frac{\omega T}{2\pi} \right) \end{aligned}$$



Frequency Domain view of ZOH



Frequency Response $|H_0(j\omega)|$



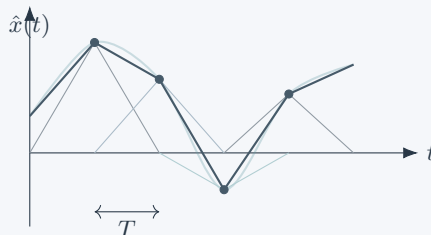
Impulse Response $h_0(t)$

- $h_0(t)$ is a rectangle of width T . Its frequency response includes a sinc function:
$$H_0(j\omega) = T e^{-j\omega T/2} \text{sinc}\left(\frac{\omega T}{2\pi}\right).$$

We notice that,

- In the passband $|\omega| \leq \omega_s/2$, the gain is not constant, attenuating high frequencies slightly.
- Outside the passband, the gain is non-zero, allowing high-frequency aliasing copies to leak through.

- **Linear Interpolation** connects adjacent samples with straight lines.
- This produces a continuous, much smoother approximation $x_1(t)$ than the ZOH staircase.
- Achieved by filtering the impulse-sampled signal $x_p(t)$ with a triangular pulse $h_1(t)$.



- The impulse response $h_1(t)$ is a triangle spanning $[-T, T]$ with peak height 1 at $t = 0$.

Why is $h_1(t)$ a Triangle?

- In a reconstructed signal, a given sample $y(nT)$ contributes via $y(nT) \cdot h_1(t - nT)$.
- The shape of $h_1(t)$ thus dictates the **weighting** of $y(nT)$ over time.

For Linear Interpolation:

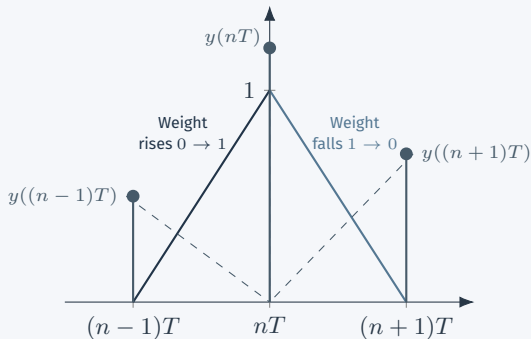
Given a weight $\alpha \in [0, 1]$, we should have,

$$y(t) = (1 - \alpha)y((n - 1)T) + \alpha y(nT) \quad \text{for } t \in [(n - 1)T, nT]$$

$$y(t) = (1 - \alpha)y(nT) + \alpha y((n + 1)T) \quad \text{for } t \in [nT, (n + 1)T]$$

Therefore,

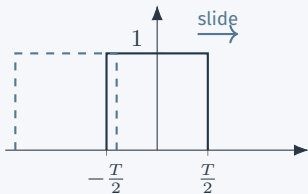
- The weight of $y(nT)$ should linearly rise strictly from 0 to 1 between $(n - 1)T$ and nT .
- It should linearly fall from 1 back to 0 between nT and $(n + 1)T$.
- At all other times, its weight must precisely be 0 to isolate its exact local influence.
- Therefore, $h_1(t)$ **must** inherently be a symmetric triangle!



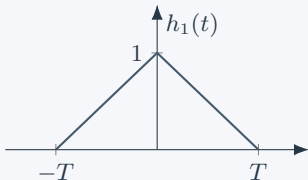
A linear interpolation triangle filter $h_1(t)$ can be formed by convolving two shifted zero-order hold rectangles!

$$h_1(t) = \frac{1}{T} \left(h_0(t + T/2) * h_0(t + T/2) \right)$$

Rectangles overlapping

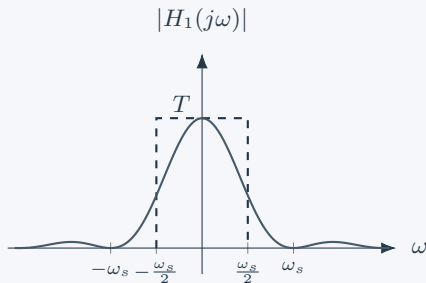


- $h_0(t + T/2)$ is just the standard ZOH block, centered exactly on time $t = 0$.
- Area of overlap grows linearly to a peak, and drops cleanly linearly!
- The peak area is $T \times 1 = T$, hence the factor $\frac{1}{T}$, as the triangle peak should be 1.

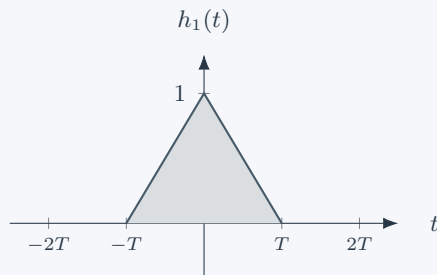


Frequency Derivation: Use $\mathcal{F}\{x * x\} = \mathcal{F}\{x\}^2$.

$$\begin{aligned} H_1(j\omega) &= \frac{1}{T} \left[\mathcal{F}\{h_0(t + T/2)\} \right]^2 \\ &= \frac{1}{T} \left[T \operatorname{sinc}\left(\frac{\omega T}{2\pi}\right) \right]^2 \\ &= T \operatorname{sinc}^2\left(\frac{\omega T}{2\pi}\right) \end{aligned}$$



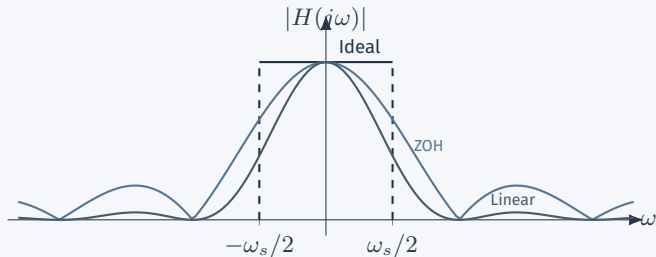
Frequency Response $|H_1(j\omega)|$



Impulse Response $h_1(t)$

- $h_1(t)$ is a triangular pulse spanning $[-T, T]$. Its frequency response is $H_1(j\omega) = T \text{sinc}^2\left(\frac{\omega T}{2\pi}\right)$.
- Because of the convolution property, $H_1(j\omega)$ is proportional to the square of the ZOH response, without the phase delay.

- Let's compare the frequency domain magnitudes against the ideal filter (assuming $T = 1$):



- Linear interpolation has much faster high-frequency rolloff (less leakage) than ZOH, making it a "better" low-pass filter approximation.

Connection to the DFT

The Uncertainty Principle

Signals **cannot** be both time-limited and strictly band-limited!

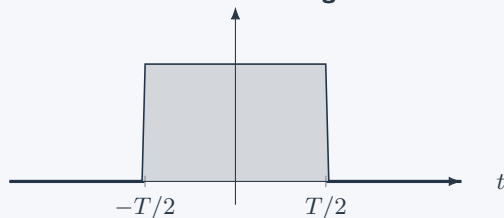
- Finite-duration captures mean $X(j\omega)$ inherently extends to infinite frequencies.^a

^aInteresting watch: 3Blue1Brown - Uncertainty Principle

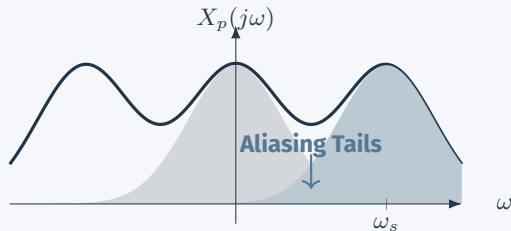
Unavoidable Aliasing

Sampling any real-world signal causes some aliasing! The infinite spectral tails will always overlap and leak into our baseband.

Time-Limited Signal



Infinite Frequency Spectrum & Aliasing



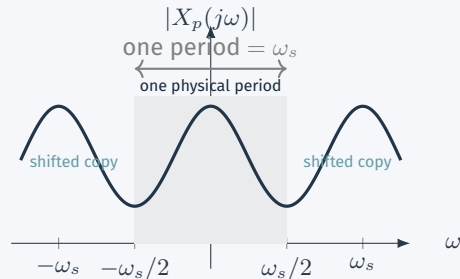
Earlier in the Nyquist proof we derived the CTFT of the sampled impulse train:

$$X_p(j\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega nT}$$

$$\begin{aligned} X_p(j(\omega + \omega_s)) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j(\omega + \omega_s)nT} \\ &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega nT} e^{-j2\pi n} \\ &= X_p(j\omega) \end{aligned}$$

- This is exactly the physical-frequency sampled spectrum from the Nyquist proof.
- It repeats every $\omega_s = \frac{2\pi}{T}$.
- Under $\omega T \mapsto \omega$, the same expression becomes the DTFT:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}.$$

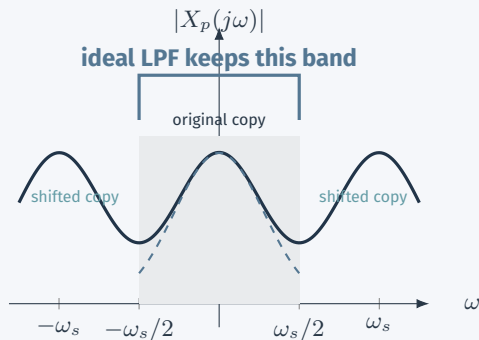


Baseband Copy, Aliasing, and the Ideal LPF

$$X_p(j\omega) = \frac{1}{T} \sum_{r=-\infty}^{\infty} X(j(\omega - r\omega_s))$$

- The central band $[-\omega_s/2, \omega_s/2]$ contains the original copy centered at 0.
- Every other interval of width ω_s is a shifted copy of that same spectrum.
- For finite-duration captures the tails are infinite, so some aliasing leakage remains, but the baseband picture is still the same one from the Nyquist proof.
- The ideal reconstruction filter keeps exactly that baseband copy:

$$H(j\omega) = \begin{cases} T, & |\omega| < \omega_s/2 \\ 0, & \text{otherwise} \end{cases}$$



- Ideal reconstruction uses one ω_s -wide period of the sampled spectrum:

$$x(t) = \frac{1}{2\pi} \int_{-\omega_s/2}^{\omega_s/2} T X_p(j\omega) e^{j\omega t} d\omega$$

- At the sample locations,

$$x[n] = x(nT).$$

- After the same normalization, we simply read this as the discrete-time sequence.

$$\begin{aligned} x[n] = x(nT) &= \frac{1}{2\pi} \int_{-\omega_s/2}^{\omega_s/2} T X_p(j\omega) e^{j\omega nT} d\omega \\ &\xrightarrow{\omega T \mapsto \omega} \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \end{aligned}$$

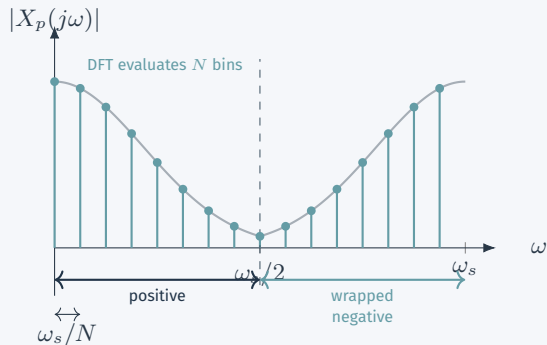
Inverse DTFT

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$[-\omega_s/2, \omega_s/2] \xrightarrow{\omega T \mapsto \omega} [-\pi, \pi]$$

$$X[k] = X_p\left(j\frac{k\omega_s}{N}\right) = X(e^{j\omega})\Big|_{\omega=\frac{2\pi}{N}k}$$

- Undoing the relabeling, a normalized DTFT point at ω sits at the physical frequency $\omega/T = \omega\omega_s/(2\pi)$.
- So an N -point DFT takes N uniformly spaced samples across one physical period of width ω_s .
- For even N , bins $k = 0, \dots, N/2$ run from dc to the highest positive frequency.
- Bins $k = N/2 + 1, \dots, N - 1$ then continue as wrapped negative frequencies: large negative first, then back toward dc.



Periodic Signals Give Harmonic Line Spectra

- If the underlying signal has period T_0 , then $\omega_0 = 2\pi/T_0$ and

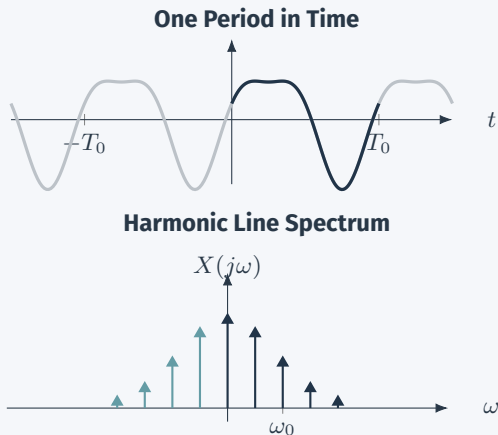
$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}.$$

- Its CTFT is therefore a line spectrum:

$$X(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0).$$

- If we sample exactly N times in one period, then

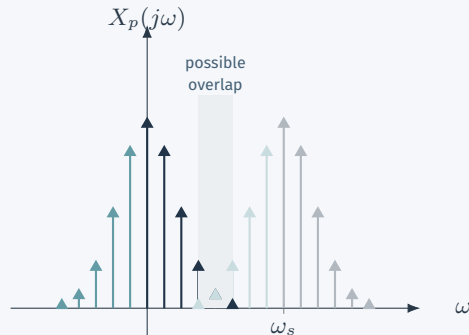
$$T_0 = NT, \quad \omega_s = \frac{2\pi}{T} = N\omega_0.$$



Sampling Copies the Periodic Line Spectrum

$$\begin{aligned}X_p(j\omega) &= \frac{1}{T} \sum_{r=-\infty}^{\infty} X(j(\omega - r\omega_s)) \\&= \frac{1}{T} \sum_{r=-\infty}^{\infty} 2\pi \sum_{k=-\infty}^{\infty} a_k \delta((\omega - r\omega_s) - k\omega_0) \\&= \frac{2\pi}{T} \sum_{r=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0 - r\omega_s)\end{aligned}$$

- Each harmonic line is copied to every passband.
- Depending on the bandwidth, the copied line spectra may overlap and alias into each other.



Nyquist Condition for the Periodic Case

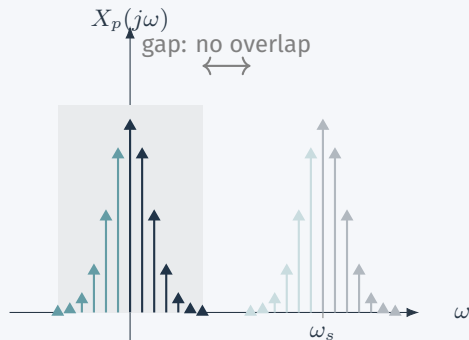
If the original periodic signal is bandlimited to the Nyquist band,

$$|k\omega_0| < \omega_s/2,$$

then $a_k = 0$ whenever $|k| \geq N/2$. The copied line spectrum becomes

$$X_p(j\omega) = \frac{2\pi}{T} \sum_{r=-\infty}^{\infty} \sum_{|k| < N/2} a_k \delta(\omega - k\omega_0 - r\omega_s).$$

- Now one passband contains one clean, non-overlapping copy of the original harmonic line spectrum.
- This is the periodic-signal version of the Nyquist condition.



The Same Spectrum from the Sampled Sequence

From the recap derivation, the sampled periodic signal itself has the Fourier series

$$x_p(t) = \sum_{\ell=-\infty}^{\infty} c_{\ell} e^{j\ell\omega_0 t}, \quad c_{\ell} = \frac{1}{T_0} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}\ell n} = \frac{1}{T_0} X[\ell].$$

Since $c_{\ell+N} = c_{\ell}$, choose any N consecutive indices $\mathcal{K}$ and write $\ell = k + rN$:

$$\begin{aligned} X_p(j\omega) &= 2\pi \sum_{\ell=-\infty}^{\infty} c_{\ell} \delta(\omega - \ell\omega_0) \\ &= 2\pi \sum_{r=-\infty}^{\infty} \sum_{k \in \mathcal{K}} c_{k+rN} \delta(\omega - (k+rN)\omega_0) \\ &= 2\pi \sum_{r=-\infty}^{\infty} \sum_{k \in \mathcal{K}} c_k \delta(\omega - k\omega_0 - r\omega_s). \end{aligned}$$

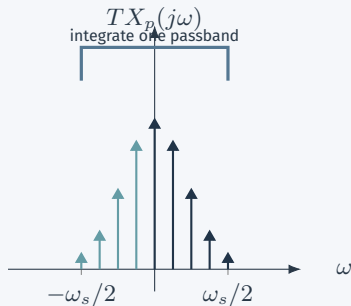


Coefficient Match

$$c_k = \frac{a_k}{T}, \quad X[k] = T_0 c_k = \frac{T_0}{T} a_k = N a_k$$

The comparison is made on the non-overlapping baseband harmonics. If $T = 1$, this reads $X[k] = T_0 a_k$.

Reconstruction Gives $x(t)$ and the IDFT



Inside one passband, only the non-overlapping baseband harmonics remain:

$$\begin{aligned}x(t) &= \frac{1}{2\pi} \int_{-\omega_s/2}^{\omega_s/2} TX_p(j\omega) e^{j\omega t} d\omega \\&= T \sum_{|k| < N/2} c_k e^{jk\omega_0 t} = \sum_{|k| < N/2} a_k e^{jk\omega_0 t}.\end{aligned}$$

Thus the ideal LPF perfectly recovers the continuous-time periodic signal whenever Nyquist holds.

Sampling at $t = nT$ and using $T_0 = NT$ gives

$$\begin{aligned}x[n] = x(nT) &= T \sum_{|k| < N/2} c_k e^{jk\omega_0 nT} \\&= \frac{1}{N} \sum_{|k| < N/2} X[k] e^{j\frac{2\pi}{N} kn}.\end{aligned}$$

Using the periodicity of $X[k]$, we can shift the index set to $k = 0, \dots, N-1$:

Inverse DFT

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N} kn}$$